Q-Weighted Politician Event Studies

Exploratory fixed-effect comparison

August 18, 2026

Connection to the paper. This exploratory analysis applies the corrective Q-weighted stacked difference-in-differences estimator from Wing, Freedman, and Hollingsworth (2024), *Stacked Difference-in-Differences*. We replicate their method and their original fixed-effect specification, not their numerical Medicaid results: the estimates below are for fires in this project. The paper's preferred fixed-composition target is a trimmed aggregate ATT.

What cohort ID means. A cohort ID labels one stacked sub-experiment (a province-election pair); it does not label a control type. Every included cohort contains both treated observations ($treat = 1$) and controls ($treat = 0$). All $treat = 0$ observations are included because they remain untreated throughout that stack's analysis window; no distinction among control types is used.

All cohorts are included. No cohort is discarded for lacking some leads or lags. Each sub-experiment contributes only at the event years available in the data. Thus IDs 3–6 contribute from -5 through 4; IDs 1–2 enter later; and IDs 7–8 leave after event year 2.

ID	Province	Election	Available event years	Used?
1	Haryana	2014-11	-3 to 4	Include
2	Bihar	2015-12	-4 to 4	Include
3	Punjab	2017-04	-5 to 4	Include
4	Uttar Pradesh	2017-04	-5 to 4	Include
5	Haryana	2019-11	-5 to 4	Include
6	Bihar	2020-12	-5 to 4	Include
7	Punjab	2022-04	-5 to 2	Include
8	Uttar Pradesh	2022-04	-5 to 2	Include

Corrective weights. For cohort s and event year e, treated observations receive weight one and controls receive

$$Q_{se} = \frac{n_{se}^T/N_e^T}{n_{se}^C/N_e^C}.$$

At each event year, this makes the weighted cohort shares among controls match the shares among treated observations for the cohorts available at that event year, which is the paper's corrective-weight principle.

Estimand caveat. Because cohort availability changes over event time, this requested all-cohort specification is not the paper's fixed-composition trimmed aggregate ATT over -5 through 4. Event year -5 uses IDs 3–8; -4 uses IDs 2–8; -3 through 2 use IDs 1–8; and 3–4 use IDs 1–6. Consequently, differences across event years can reflect both dynamic effects and changes in the contributing cohorts.

Event-study estimates

Event years run from -5 through 4, with -1 omitted. The sample contains 12,599,763 observations from cohort IDs 1, 2, 3, 4, 5, 6, 7, 8. All models include wind direction and average wind speed, use Q-weights, and cluster standard errors by AC × cohort. Control weights range from 0.167 to 1.938.

Specifications. Original stacked FE absorbs treatment-group and relative-year effects. FE 01 absorbs grid × cohort effects. FE 05 absorbs grid × cohort and province × cohort × election-year effects.

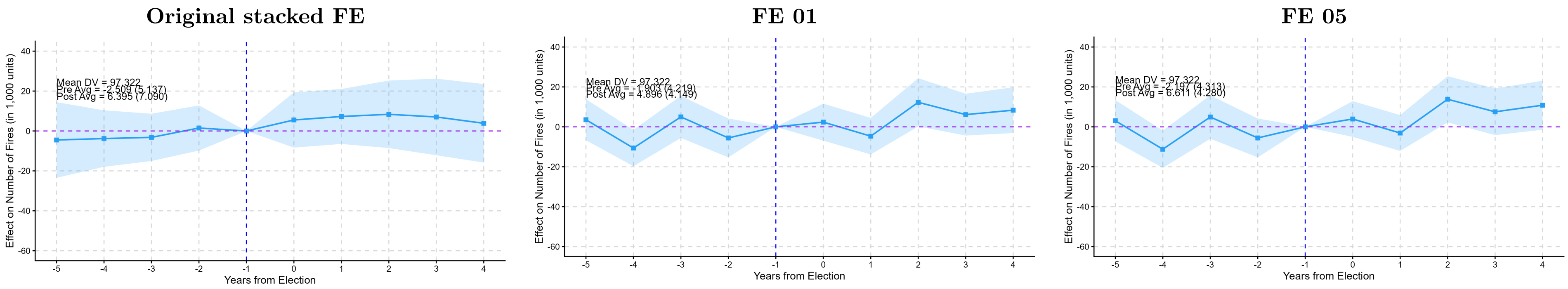

Figure 1: Q-weighted event-study estimates and 95% confidence intervals.

Q-weighted DiD interactions

The two post-treatment estimates are intentionally offset horizontally. Their coefficient values and confidence intervals are unchanged; the added space only makes overlapping 90% and 95% intervals visible.

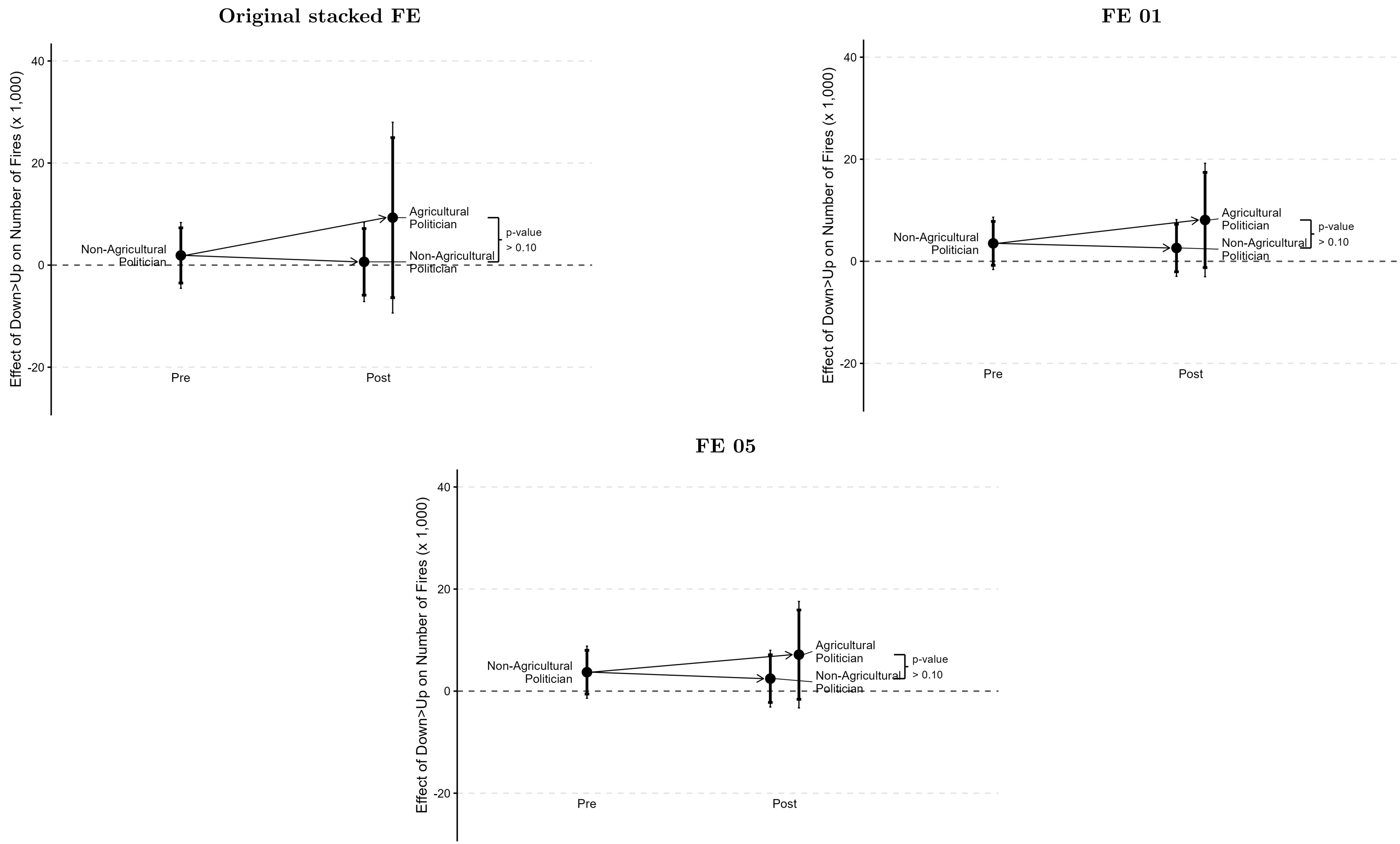

Figure 2: Q-weighted pre/post interactions with separated post positions.

Regression estimates

Dependent Variable:	countk		
Specification	Original stacked FE	FE 01	FE 05
Model:	(1)	(2)	(3)
Variables			
treat × relative_year = -5	-4.52	3.58	3.04
	(9.67)	(5.24)	(5.28)
treat × relative_year = -4	-3.80	-10.6**	-11.2**
	(7.22)	(4.63)	(4.76)
treat × relative_year = -3	-3.20	4.99	4.93
	(6.04)	(5.38)	(5.54)
treat × relative_year = -2	1.48	-5.55	-5.55
	(5.76)	(4.96)	(4.96)
treat × relative_year = 0	5.51	2.37	3.94
	(7.05)	(4.73)	(4.56)
treat × relative_year = 1	7.25	-4.67	-3.10
	(7.02)	(4.66)	(4.58)
treat × relative_year = 2	8.35	12.3**	13.8**
	(8.62)	(6.16)	(5.89)
treat × relative_year = 3	7.02	6.13	7.53
	(9.78)	(5.34)	(5.96)
treat × relative_year = 4	3.85	8.35	10.8*
	(10.0)	(5.84)	(6.25)
Fixed-effects			
treat	Yes		
relative_year	Yes		
grid_cohort		Yes	Yes
province-cohort_id-election_year			Yes
Fit statistics			
Observations	12,599,763	12,599,763	12,599,763
R^2	0.00456	0.12969	0.13015
Within R^2	0.00441	0.00487	0.00485

Clustered (cluster_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The omitted event year is -1. The dependent variable is the project-standard fire count multiplied by 1,000. The same complete-case, all-cohort sample and Q-weights are used in all three columns. Cohort composition varies with event-time coverage. Method: Wing, Freedman, and Hollingsworth (2024). Implementation reference: authors' example code.